

Your design problems

Describe how you could solve each of the following problems using data structures we've learned so far. Specify the ADTs (List, Stack, Queue, Iterator, Set, Map, Priority Queue) you would use and how to use them to efficiently support the required operations.

1. You are managing a Fantasy Knox College Soccer League where students can create their own teams by selecting real players from the Knox College men's soccer team. Each student builds a fantasy team by choosing players from the official men's soccer roster. Throughout the season, players earn points based on match performances such as scoring goals, making assists, keeping clean sheets (when goalkeepers or defenders don't concede any goals) or making saves. To begin your system, you need to return a participant's roster based on their participant ID. What ADT(s) would you use to implement this functionality?
2. In a futuristic city, we have aliens that frequently visit the Earth from their planets. We use the Interplanetary Identification System (IIS) to verify each alien's identity. Every alien is identified by a unique ID (a string like "ZQ-8G9D"), along with their planet of origin (string) and birthday (string). The IIS must allow new aliens to be registered (added to the system) based on this information, existing aliens to be removed when necessary, and provide a way to find an alien's information by searching for their unique ID. What ADT(s) would you use to implement this system in the most efficient way possible?
3. In *Magic: The Gathering* players play as wizards who cast spells. When each spell is cast, but before its effects happen, the other players have a chance to respond and cast their own spell, which the other players can respond to, and so on and so forth. Once no player responds to a spell, the effects of all the spells are applied, starting with the latest spell and working backwards chronologically. What data structure(s) should be used to store the spells that have been cast, and resolve them in the correct order?
4. Sephora is a popular beauty and cosmetics retailer that sells makeup, skincare, haircare, and other beauty products both in physical stores and online. Many customers shop through Sephora's website, placing orders for products such as "Too Faced Foundation" or "Fenty Lip Gloss". Imagine you are creating a simple application to help customers keep track of their online Sephora orders. Each order has a unique ID (like "ORD123") and includes a list of products the customer purchased. The system will create a record of the order when it is placed. Then customers will be able to look up the order information using its ID. What ADT(s) would be best to implement this system?
5. You are simulating a game of Hangman. In this game, players must guess letters to spell out a hidden word. Once they guess a letter, it will be revealed where it is in the word if it is present, or, if it is not present within the word, it will be added to a pool of incorrect guesses, and another line will be drawn in a picture. If a player attempts to guess a letter that has already been guessed, they will be asked to guess again with no penalty. What ADT(s) would you use to store specifically the incorrect guesses?
6. You work for the Knox College scene shop. The scene shop holds hundreds of lighting instruments of different types (source 4, parnell, scoops). Each type of instrument has different parameters (the source 4s have different degree barrels, the parnells have different colored gels changing the output of the light, the scoops have no parameters, ect). Each light has an ID assigned to it (H-47 for example), and that is how we tell them apart. The user wants to look up the instruments by the ID, and get the type of light, and what might be attached to the light (type of lens, type of gel). What ADT(s) would you use to implement this?
7. A public library wants to create a book hold function where customers can place a book on hold that is currently checked out, and as soon as it is returned, it goes to the next customer in line. The functions that need to be included are:

- placeHold that records a customer wants a certain book,
- cancelHold that removes a customer from the waitlist, and
- seeNextCustomer that returns the customer that is next in line for the book

Each of these takes the title of the book whose record is being examined and the first two also take the customer information.

8. You have been asked to create a playlist feature system by a popular music streaming company for its music application, enabling its users to create new playlists and customize them. Each playlist should maintain an ordered list of songs, each represented as words (such as song title). The system must support the following operations for each playlist: add a song to the end of the list, insert a song at a given position, remove a song by title, and a "Next" function that allows the user to play the next song one at a time. Users should also be able to loop through the playlist in order and reset it once they reach the end of the playlist. What ADT(s) would you use to implement the playlist data structure?
9. Knox College finally hired people to deliver foods and groceries in Gizmo and C-store, and you are required to write a code to assign students a job. The order of the job has to be first-come first-serve. It means that a delivery worker has to deliver the package in the order they come in, even if there is no tip and it is just one toothpaste. A delivery worker has to know a student's ID, name, location, and items. What ADT(s) would you use to implement this system?
10. The SMC building has a lot of rooms that students can reserve throughout the day. Each reservation includes a start time, end time, student name, ID number, and the room name. The scheduler must allow the following operations:
 - Add a reservation if the room is available during the requested time.
 - Cancel a reservation, identifying it by room number, student name, ID number, and time.
 - Retrieve a list of all reservations for a specific room, sorted by start time.

What ADT(s) would you use to implement these operations?

11. It is the end of the school year and students need to choose their residence hall for next year. These are chosen based on their class year: first come last year students, then juniors, then sophomores, and then first years. If students have the same class year, they are ordered by when they completed the housing form. What ADT(s) would you use to keep track of all the students and print their names in the desired order?
12. A library is working on creating a new digital catalog for their collection. They will create a listing for each book that has the book's title, author, genre summary/description. People should be able to look up the book by its title. What ADT(s) would be used to implement this?
13. You have a shoe rack that can hold exactly 8 shoes at a time. However, you own 9 shoes in total. One shoe is always on your feet, while the other 8 are stored in the rack. Every time you come home and pick a shoe from any specific position on the rack (positions 0 through 7), the shoe you are currently wearing gets placed in that position on the rack, and the newly picked shoe becomes the one you are now wearing. You need to design a system that does the following: (a) Initializes the rack with 8 shoes and one worn shoe, (b) Let's you pick a shoe from a specific index and replaces it with the currently worn shoe, (c) Always knows which shoe is currently worn and (d) Reports which shoe was picked and which shoe was placed into the rack when you change shoes. What ADT(s) would you use to implement this functionality?